xp research

May 2023
Macro Research

# Brazil Macro Monthly

## In pursuit of additional revenue

- The global economy should decelerate amid restrictive financial conditions. Central banks have reached or are close to the end of their tightening cycle, although rates are expected to remain restrictive for some time;
- In Brazil, economic activity has surprised positively in recent months. As a result, we raised our forecast for 2023 GDP growth, to 1.4% from 1.0%. In turn, our expectation for 2024 GDP growth remains at 1.0%;
- Services inflation remains elevated and expansionary fiscal policy should keep inflation under pressure ahead. We keep our forecasts of 6.2% for 2023 and 5.0% for 2024;
- The new fiscal rule should not generate short-term improvements. Even with additional revenues, we expect a deficit of 1.1% in 2023 and 1.0% in 2024 and public debt to continue growing;
- The Brazilian exchange rate has strengthened in recent weeks, mostly driven by external factors. Thus, we assign a downward bias to our current forecast of BRL/USD 5.30 for YE 2023;
- We believe the Copom will start a gradual easing cycle in the second semester. Since fiscal policy will likely be expansionary ahead, the Selic rate tends to be higher in the steady state. We forecast 11.00% for 2024.


**Caio Megale**
Chief economist

**Andres Pardo**
Chief Latam Strategist

**Rodolfo Margato**
Economist

**Tiago Sbardelotto**
Economist

**Alexandre Maluf**
Economist

**Francisco Nobre**
Economist

**Alberto Silva**
Intern

**Contact**
economiaxp@xpi.com.br


XP Forecasts

| | 2019 | 2020 | 2021 | 2022 | 2023 (F) | 2024 (F) |
|---|---|---|---|---|---|---|
| GDP growth (%) | 1.2 | -3.3 | 5.0 | 2.9 | 1.4 | 1.0 |
| Unemployment rate (% s.a., end of period) | 11.5 | 14.7 | 11.7 | 8.3 | 9.2 | 9.8 |
| IPCA (CPI, 12m, %) | 4.3 | 4.5 | 10.1 | 5.8 | 6.2 | 5.0 |
| SELIC rate (% p.y., end of period) | 4.50 | 2.00 | 9.25 | 13.75 | 12.00 | 11.00 |
| FX (USDBRL, end of period) | 4.03 | 5.20 | 5.58 | 5.28 | 5.30 | 5.40 |
| Central government primary balance (% GDP) | -1.2 | -9.8 | -0.4 | 0.5 | -1.2 | -1.0 |
| Public sector primary balance (% GDP) | -1.0 | -9.3 | 0.7 | 1.2 | -0.9 | -0.8 |
| Gross debt (% GDP) | 74.3 | 86.9 | 78.8 | 73.5 | 75.9 | 78.7 |
| Trade Balance (USD Bn) | 35.2 | 50.4 | 61.4 | 61.7 | 73.0 | 64.5 |
| Exports (USD Bn) | 221.1 | 209.2 | 280.8 | 334.4 | 360.5 | 358.1 |
| Imports (USD Bn) | 185.9 | 158.8 | 219.4 | 272.7 | 287.5 | 293.6 |
| Current Account (USD Bn) | -68.0 | -28.2 | -46.4 | -55.7 | -40.0 | -48.0 |
| Current Account (% GDP) | -3.6 | -1.9 | -2.8 | -2.9 | -1.9 | -2.3 |
| FDI (USD Bn) | 69.2 | 37.8 | 46.4 | 90.6 | 80.0 | 75.0 |
| FDI (% GDP) | 3.7 | 2.6 | 2.8 | 4.8 | 3.9 | 3.6 |
| Nominal GDP (USD Tri) | 1.873 | 1.475 | 1.649 | 1.919 | 2.065 | 2.108 |
| Nominal GDP (BRL Tri) | 7.389 | 7.610 | 8.899 | 9.915 | 10.637 | 11.276 |

Source: IBGE, BCB, Bloomberg, XP.

# Foreword – Fiscal challenges remain amid still benign global outlook

Despite the uncertainties, particularly in the US banking system, we see the global economy getting more becoming for Brazil. US and Europe will probably not hike rates much further. Recession, if controlled, tends do be positive news, as it helps to bring inflation down (bear in mind that Brazil has quite a limited trade flow with developed economies). China and India, in its turn, are recovering, which should provide support for soft commodity and energy prices.

Domestically, though, challenges continue. The government sent the new fiscal framework to Congress, but the bill lacks the enforcement needed to balance the budget. Minister Haddad has emphasized that the government will increase revenues by reducing tax benefits for the rich. It is not clear if he will succeed. We believe, *ceteris paribus*, tax collections will increase somewhat, though not sufficient to stabilize debt-to-GDP dynamics.

But we are not in a "ceteris paribus" world. Economic activity is losing steam – though by less than expected - which has already hit tax collections. Indeed, recent fiscal results came out below expectations. This means that if the new fiscal framework passes Congress without changes, fiscal policy will put pressure on inflation in the years to come.

Thus, we see only limited room for the central bank to ease monetary policy in the coming quarters. As global inflation is falling and domestic activity is decelerating, there is no need for monetary policy to remain at the same tight level it was at the beginning of the year (around 8.0% real interest rates). So, we continue to see the BCB starting a gradual easing cycle in August, but with a terminal Selic rate still at double digits - 11.0% by the first semester of 2024.

Against this backdrop, we anticipate that tensions between monetary and fiscal policy will continue going forward. Whether it will reflect on Brazilian assets' risk premium will depend on whether the global outlook continues to be benign, as described above.

# Global backdrop: Recession in developed economies becoming increasingly likely

**Tail risks persist, although a mind recession in the US is the base case**. The risk of a "no landing" scenario in which the US economy would not decelerate, and inflation would not converge to the target seems increasingly unlikely. Activity indicators have been deteriorating, pointing to below-trend growth, and are consistent with a recession in the US. Meanwhile, concerns of a financial crisis persist, although it does not appear to be a systematic issue. Our base case scenario suggests that inflation dynamics converge to the 2-3% range amid a moderate recession. Recession risk and stabilizing monetary policy have partially reverted the USD´s recent appreciation (graph).

**We believe the Fed has paused the tightening cycle.** The Fed hiked rates by 25 bps in May, leaving doors open for upcoming meetings. This decision took the upper bound of the Fed Funds rate to 5.25%, which we consider sufficiently restrictive to guarantee the convergence of inflation. Despite market pricing suggesting rate cuts in the second half of 2023, we believe authorities are unlikely to ease monetary policy this year. Core inflation has remained significantly pressures, especially on the services side, and the labor market remains tight. These developments will likely keep the Fed hawkish for some time. We expect the FOMC to gradually start cutting rates in 1Q24, ending the year with the Fed Funds rate at 3.5%. Real interest rates will likely return to its neutral level in 2025 (around 0.5%), although the nominal level will largely depend on where inflation stabilizes.

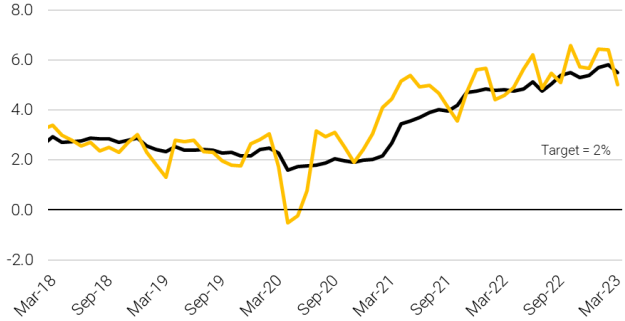
**The ECB will stay hawkish for longer.** The ECB has continued its tightening cycle by increasing its three key interest rates by 25 bps. Cor inflation in the Eurozone has shown no relief yet, keeping policymakers hawkish. ECB officials have said they expect to continue tightening monetary policy until inflation dynamics show clear signs of convergence toward the 2% target. We expect the ECB to hike another 50-75 bps before ending the tightening cycle, and consequently, European economies should weaken considerably going forward.

**China presents stronger than expected growth in Q1.** GDP expanded by 4.5% y/y in 1Q23 and exceeded analysts' expectations. The strong performance was driven by solid expansion in exports and infrastructure investment and a rebound in property prices. 1Q23 results showed a material improvement from China's average GDP growth rate of 3.0% in 2022, although it still fell short of the 5% growth target set for 2023. Going forward, we expect the Chinese economy to continue accelerating, mainly driven by a strong recovery in consumption. Household accumulated savings are at historically high levels, while consumer confidence has not yet recovered since the reopening. We expect China's economy to grow by 5.6% this year.

The USD weakened recently
Fred Nominal Broad U.S. Dollar Index

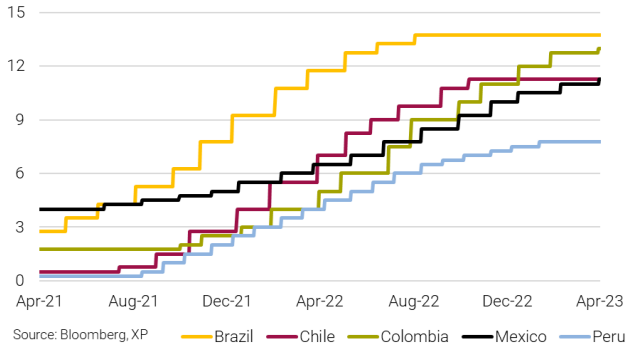
Source: St. Louis Fed, XP

Services inflation still pressured in the US
Services PCE deflator - % ch.




Source: BBG, BEA, XP

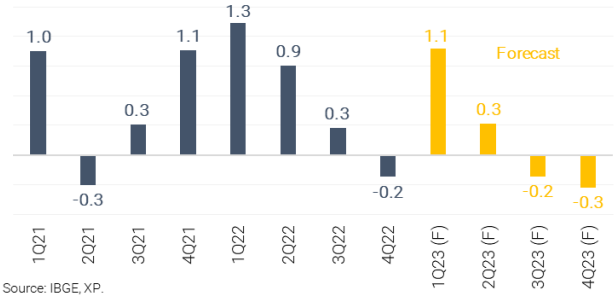
Interest rates in Latin America are expected to remain restrictive for a long time
Annual policy rate (%)




Source: Bloomberg, XP

**Latam: tightening cycles are ending, but easing should take a while.** Rate cuts are expected at different times later this year or early in 2024, but the path toward more normal rates (neutrality) will likely be longer than expected amid persistently high inflation. Brazil, Chile, and Peru already ended their tightening cycle last year or earlier this one (in the case of Peru), while Mexico and Colombia are close to reaching their terminal rates. Also, different points in the business cycle and the disinflation process will likely make central banks start cutting rates at different times. Uruguay was the first country in the region to begin cutting rates in a cautious manner. Brazil was one of the first in the region to begin raising rates and should find room to start cutting in Q3, also cautiously. We believe Chile and Peru will be the next ones to kick off their own easing cycles in 3Q23, while Colombia and Mexico may have to wait a bit longer.

We forecast strong GDP growth in 1Q (boosted by Agriculture) and gradual cooling ahead
% QoQ s.a % ch.



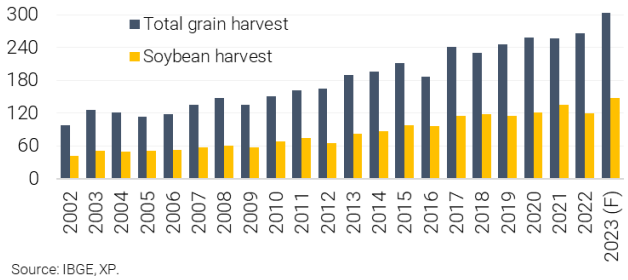

Source: IBGE, XP.

# Brazil: In pursuit of additional revenue

## Economic Activity – We raised our forecast for 2023 GDP growth, to 1.4% from 1.0%

**Activity indicators have surprised positively in recent months.** The labor market showed resilience in early 2023, with emphasis on the significant net addition of formal jobs (around 430k in 1Q, discounting seasonal effects). Furthermore, service activities have grown above expectations, while agricultural production estimates have improved further. Our updated projection for 1Q23 GDP growth stands at 1.1% QoQ (2.9% YoY) – we were anticipating an increase of 0.8% a month earlier. This expectation is mainly due to the strong recovery of Agriculture GDP, boosted by the soybean crop. Total GDP would grow by 0.3% QoQ if it were not for the contribution of the primary sector. 1Q23 GDP results will be published on June 1st.

Grain harvest is expected to surge nearly 15% in 2023, a new all-time high
(in millions of tons)



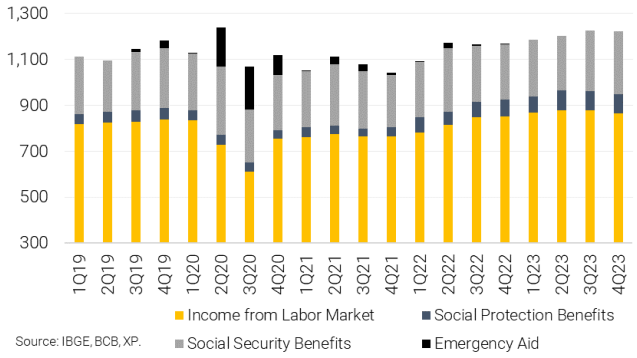

Source: IBGE, XP.

**Household consumption and exports continue to play a major role.** As we argued in our latest monthly report, real household disposable income will rise substantially in 2023 (4.2% as per our forecasts), on the heels of rising wages and higher government cash transfers. Regarding the latter, the expansion of the "Bolsa-Família" welfare program is the highlight, given the amount of nearly BRL 170 billion this year. Income impulses will soften the slowdown in household consumption going forward. Moreover, Brazilian exports are expected to increase sharply, owing to the solid external demand and high commodity prices. In this context, we call attention to the material contribution from sectors less sensitive to the economic cycle, such as the agriculture and mining industries.

**Therefore, we revised upwards our projection for 2023 GDP growth, to 1.4% from 1%.**

**We continue to anticipate a weakening of domestic activity throughout the year.** The positive surprises in recent months do not change our view that the economy will lose momentum in coming quarters. Financial conditions remain tight, while the indebtedness of households and companies has risen

Household disposable income is expected to grow by around 4% and support consumption this year
BRL billion, adjusted by inflation, s.a.s.




Source: IBGE, BCB, XP.

considerably in recent months. Corporate credit concessions have been deteriorating, especially among small and medium-sized businesses. Accordingly, private investments should decline in 2023. For instance, the latest data on the production of capital goods and civil construction came in below expectations.

**Our projection for 2024 GDP growth remains at 1.0%.** The outlook for economic activity next year is full of uncertainties. In the domestic environment, there are many doubts about the magnitude of the fiscal impulse and the room for interest rate cuts. On the international front, especially in the US and Europe, opinions are increasingly divided between soft landing and hard landing in economic activity. Our baseline scenario foresees moderate growth for Brazil's GDP in 2024 (1% increase), with a slightly negative carry-over effect and an average quarterly growth of 0.4% QoQ.

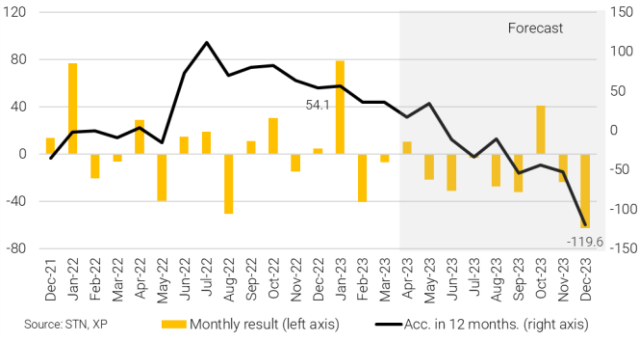
| GDP Breakdown (annual change) | 2020 | 2021 | 2022 | 2023 (F) | 2024 (F) |
|---|---|---|---|---|---|
| Agriculture and Livestock | 4.2 | 0.3 | -1.7 | 8.5 | 2.0 |
| Industry | -3.0 | 4.8 | 1.6 | 0.2 | 0.7 |
| Services | -3.7 | 5.2 | 4.2 | 1.2 | 0.9 |
| **Total GDP** | **-3.3** | **5.0** | **2.9** | **1.4** | **1.0** |
| Household Consumption | -4.6 | 3.7 | 4.3 | 2.2 | 0.9 |
| Government Consumption | -3.7 | 3.5 | 1.5 | 0.8 | 0.7 |
| Gross Fixed Capital Formation | -1.7 | 16.5 | 0.9 | -2.4 | 0.5 |
| Exports | -2.3 | 5.9 | 5.5 | 3.0 | 1.2 |
| Imports (-) | -9.5 | 12.0 | 0.8 | 1.3 | 1.0 |

Source: IBGE, XP.

## Fiscal Policy – Fiscal balance to remain negative even with additional revenues

**Recent data show a deterioration in fiscal results**. Federal tax collection continues to slow down, driven by tax cuts implemented last year, the fall in international commodity prices and a cooling economic activity. On the expenditure side, there was a significant increase in the Bolsa-Familia cash-transfer program and discretionary expenses. In 12 months, the public sector surplus dropped from BLR 93.2 billion (0.9%) in February to BRL 74.8 billion (0.7%) in March.

We expect the central government to show primary deficits in the coming months




**We project the negative trend to continue**. Revenues should improve with recent government measures, but not to the point of reversing the recent slowdown trend. Expenses, driven by the Transition PEC, should grow by almost 7.0% in real terms compared to last year. **Thus, we maintain our forecast of a BRL 119.6 billion deficit for 2023 (-1.1% of GDP).**

**Symmetric risks ahead.** The effectiveness of government measures on revenues is still uncertain since some of them need to be discussed in the judiciary and/or in Congress. The final impact can be significantly higher or lower than we currently estimate. On the expenditure side, doubts remain regarding the "speed limit" for next year (that will be established by the new fiscal framework) and the definitions of minimum-wage increase and civil servants' salaries policies.

**The new fiscal rule was formally presented to Congress**. The proposal combines a real spending growth limit between 0.6% and 2.5% and primary balance targets. As it was presented, the primary balance targets are not binding constraints for expenditures to grow. It reduces the effectiveness of the rule as a fiscal anchor for the medium term.

Measures annonuced by government

| Measures | Gross Revenue Estimates (BRL billion) Government | XP |
|---|---|---|
| Online betting taxation | 12 | 6 |
| Costing subsidy on ICMS | 80 | 32 |
| E-commerce taxation | 7 | 2 |
| **Total** | **99** | **40** |

**Fiscal forecasts for 2024 depend on the definition of expenditure growth...** as mentioned above, the limit will be corrected by this year's inflation plus a portion of the variation in revenues. Considering the possible lower (0.6%) and upper (2.5%) limits and our revenue scenario, the primary result could be between BRL 110.7 billion (-1.0% of GDP) and BRL 149.5 billion (-1.3% of GDP).

**...And the government's success in raising revenue**. To reach a neutral result in 2024 that accomplishes the primary balance target established in the Budget Guidelines Law, the government needs between BRL 110 and BRL 150 billion in additional net revenue. However, the measures announced by the government so far are insufficient to achieve this figure.

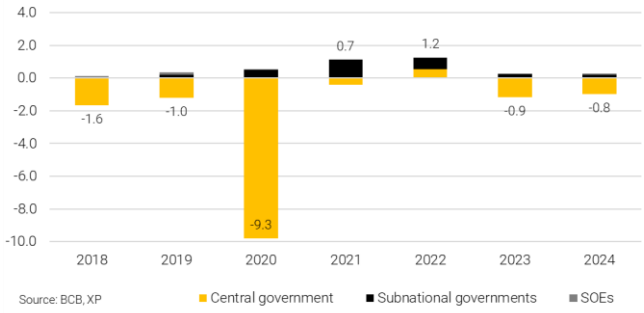
**Thus, we project a deficit of BRL 109.1 billion (1.0% of GDP) in 2024**. In our base scenario, we consider an expenditure growth of 1.6% above inflation and additional gross revenues of BRL 30 billion (net revenues BRL 28 billion) for the central government.

**For the public sector, we estimate a deficit of BRL 99.4 billion in 2023 and BRL 80.6 billion in 2024**. Our estimate for subnational governments considers the partial recovery of ICMS (state-level VAT) revenues with the change in gasoline and ethanol taxes as of June and the change in the tax base on electricity.

**Public debt on the rise**. After a mild drop in January, the general government's gross debt rose back to 73.0% in March. We expect the debt to continue an upward trend throughout this and next year. It is important to point out that, in the absence of additional measures to increase collections in the coming years, we do not foresee debt stabilization.

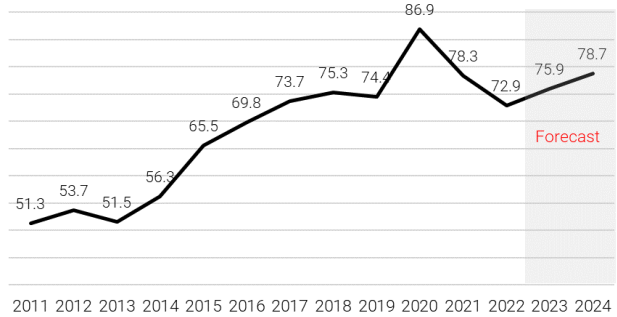
**Fiscal rule and tax reform are key fiscal themes for Congress**. Our base hypothesis is that the proposal for a new fiscal rule will be approved with minor adjustments. The discussion on the tax reform should start next, but its progress is uncertain given the pressures from companies and sectors. The approval of the proposals is essential for sustainable and stronger growth ahead.

Despite the new fiscal rule, public sector primary balance should be negative in 2023 and 2024
Public sector balance - as % of GDP




Source: BCB, XP

Public debt to grow even with lower interest rates
General government gross debt - as % of GDP




Source: BCB, XP

## Inflation – Moderate (and incomplete) disinflation throughout 2024

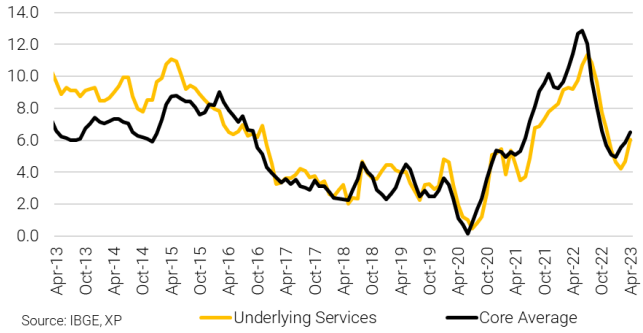
**Deceleration of annual inflation (mainly) due to the base effect.** The March IPCA cooled down to 4.65% year-on-year (compared to 5.60% in February), below our estimates and market consensus. The drop is related to the base effect (tax cuts during the election period) and global cost deflation. Core measures and services inflation remains under pressure.

**Underlying services, the subgroup most linked to economic activity and labor-intensive items, remained stable at 5.0% (3M SAAR)**. In fact, the April mid-month consumer inflation IPCA-15 showed an acceleration in this group. For now, it is hard to say whether this indicates a trend change or a temporary rise.

In fact, economic activity and the labor market have been more resilient than expected, in line with our upward revision for the 2023 GDP (see the activity section above). As of now, we project a slight deceleration in the services metric from 7.6% in March to 6.2% at the end of 2023, which increases the challenge for the monetary authority to bring inflation toward the target.

**Regarding administered prices**, we incorporated readjustments for lottery bets and the increase in bus fares in Belo Horizonte, with an upward effect of almost 0.10pp in our forecast. For the year, we marginally raised our monitored inflation forecast to

Cores and underlying services remain resistant, reflecting the tough path of disinflation in Brazil
(IPCA-15 - y/y % ch.)



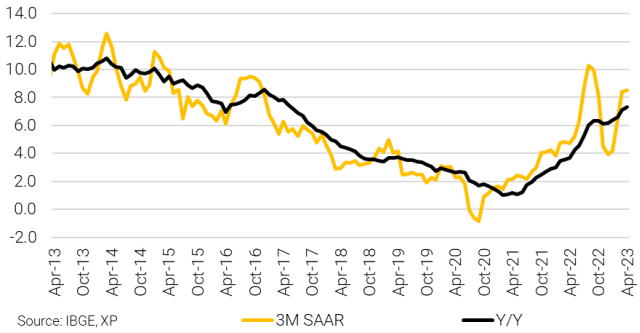

Source: IBGE, XP

**11.4%, a high level, mostly explained by the return of federal and state taxes on fuel**, in addition to a significant increase in electricity.

**Considering the abovementioned topics, we maintain our 2023 IPCA forecast at 6.2%. If the recent drop in food and oil prices persists throughout the year, the risk will become slightly tilted to the downside.**

**For 2024**, we expect inflation to **continue to decline, benefiting from the global disinflation process and the tightening cycle by the BCB**. However, the still **expansionary fiscal policy should prevent a more pronounced disinflation**, both due to its effect on aggregate demand and public debt sustainability. As we pointed out in our last Brasil Macro Monthly, we incorporated into our scenario a change in the inflation target and/or extension of the convergence horizon (with moving horizons), which should keep inflation expectations under pressure. Thus, **we maintain our expectation for the 2024 IPCA at 5.0%.**

Labor-intensive services prices reaccelerated in April, raising concerns about the inflation dynamics throughout the year
(IPCA-15 - % ch.)

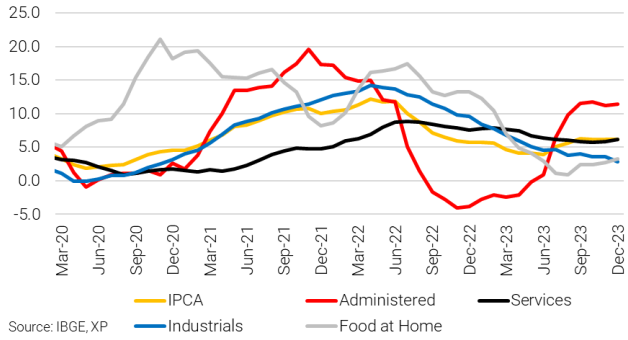



## External Sector: Global environment brings appreciation bias to the BRL

**The Brazilian exchange rate has strengthened (moderately) in recent weeks, led by external factors.** The pair fluctuated around BRL/USD 5.20 in the last quarter. In our view, the recent FX appreciation largely reflected the global weakening of the US dollar, amid still high commodity prices and the prospect of less pressured interest rates in the US. Fiscal and political uncertainties in the domestic environment seem to act in the opposite direction.

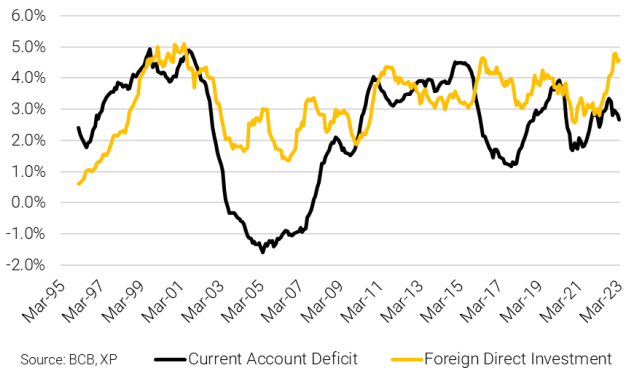
**We maintain our exchange rate forecast of BRL/USD 5.30 for YE 2023, with an annual average of BRL/USD 5.15. There is a downward bias (stronger Real) if the US dollar devaluation trend persists.**

Services and administered prices to add pressure by the end of the year
(IPCA - y/y % ch.)




**The balance of payments remains robust, with emphasis on the strong trade surplus and massive FDI (Foreign Direct Investment).** The Brazilian trade balance continues on an upward trajectory, driven by the sharp increase in exports. The grain harvest is at an all-time high, solid external demand – economic acceleration in China plays an important role – and favorable terms of trade will likely lead to a record trade surplus this year. Moreover, the deficits in the services and primary income accounts have been gradually decreasing, in line with the cooling of economic activity. Consequently, we anticipate a current account deficit of USD 40 billion at the end of 2023 (1.9% of GDP) and USD 48.0 billion at the end of 2024 (2.3% of GDP), against the USD 55.7 billion at the end of 2022 (2.9% of GDP). In turn, net inflows in FDI are expected to remain at historically high levels and record USD 80 billion this year (3.9% of GDP) and USD 75.0 billion next year (3.6% of GDP), slightly below the USD 90.6 billion (4.8% of GDP) reported last year.

FDI comfortably covers the Current Account Deficit
(acc. in 12 months - as % of GDP)



## Monetary Policy – Gradual easing in 22H2

The Copom has been keeping a hawkish tone in its recent official communication, and core inflation remains stubbornly high. Thus, we believe that the Selic rate will continue at current contractionary levels for some more time, which is consistent with Copom's signaling to keep the policy rate at 13.75% for "a long period."

At the same time, evidence is mounting that economic activity is losing steam. The unemployment rate has edged up. Credit markets are finally feeling the effects of monetary tightening. Additionally, exogenous events (such as the problems with Lojas Americanas) hit markets over and above what would be expected by tight monetary policy, particularly for SMEs.

The government's new fiscal framework does not stabilize the debt-to-GDP ratio but brings some visibility, as it sets limits for fiscal expansion ahead. Thus, the framework tends to be helpful for monetary policy management; though it does not necessarily open room for a lower interest rate over time since the fiscal stance will remain expansionary.

All considered, we believe the Copom will start a gradual easing cycle in the second semester. We see a 25-bps cut in the August Copom meeting, followed by sequential 50-bps cuts until the Selic reaches 11.00% by the first half of 2024.

Given that we believe fiscal and quasi-fiscal policies will remain expansionary in the coming years, and global neutral interest rates tend to be higher than in recent years[1], we forecast a terminal Selic rate (11.0%) that is considerably above pre-pandemics levels.

| XP Forecasts | 2019 | 2020 | 2021 | 2022 | 2023 (F) | 2024 (F) |
|---|---|---|---|---|---|---|
| GDP growth (%) | 1.2 | -3.3 | 5.0 | 2.9 | 1.4 | 1.0 |
| Unemployment rate (% s.a., end of period) | 11.5 | 14.7 | 11.7 | 8.3 | 9.2 | 9.8 |
| IPCA (CPI, 12m, %) | 4.3 | 4.5 | 10.1 | 5.8 | 6.2 | 5.0 |
| SELIC rate (% p.y., end of period) | 4.50 | 2.00 | 9.25 | 13.75 | 12.00 | 11.00 |
| FX (USDBRL, end of period) | 4.03 | 5.20 | 5.58 | 5.28 | 5.30 | 5.40 |
| Central government primary balance (% GDP) | -1.2 | -9.8 | -0.4 | 0.5 | -1.2 | -1.0 |
| Public sector primary balance (% GDP) | -1.0 | -9.3 | 0.7 | 1.2 | -0.9 | -0.8 |
| Gross debt (% GDP) | 74.3 | 86.9 | 78.8 | 73.5 | 75.9 | 78.7 |
| Trade Balance (USD Bn) | 35.2 | 50.4 | 61.4 | 61.7 | 73.0 | 64.5 |
| Exports (USD Bn) | 221.1 | 209.2 | 280.8 | 334.4 | 360.5 | 358.1 |
| Imports (USD Bn) | 185.9 | 158.8 | 219.4 | 272.7 | 287.5 | 293.6 |
| Current Account (USD Bn) | -68.0 | -28.2 | -46.4 | -55.7 | -40.0 | -48.0 |
| Current Account (% GDP) | -3.6 | -1.9 | -2.8 | -2.9 | -1.9 | -2.3 |
| FDI (USD Bn) | 69.2 | 37.8 | 46.4 | 90.6 | 80.0 | 75.0 |
| FDI (% GDP) | 3.7 | 2.6 | 2.8 | 4.8 | 3.9 | 3.6 |
| Nominal GDP (USD Tri) | 1.873 | 1.475 | 1.649 | 1.919 | 2.065 | 2.108 |
| Nominal GDP (BRL Tri) | 7.389 | 7.610 | 8.899 | 9.915 | 10.637 | 11.276 |

Source: IBGE, BCB, Bloomberg, XP.

[1] IMF World Economic Outlook, April 2023

# Disclaimer

1) This report was prepared by XP Investimentos CCTVM S.A. ("XP Investimentos or XP") according to the requirements provided in CVM Resolution 20/2021 and aims to provide information that can help the investors make their own investment decisions, and does not constitute any kind of offer or purchase request and/or sale of any product. The information contained in this report is considered valid on the date of disclosure and has been obtained from public sources. XP Investimentos is not liable for any decisions made by the customer based on this report.

2) This report was prepared considering the product risk classification in order to generate allocation results for each investor profile.

3) All of the views expressed in this research report accurately reflect the research analyst's personal views regarding any and all of the subject securities or issuers. No part of analyst(s) compensation was, is or will be, directly or indirectly related to the specific recommendations or views expressed in this research report.

4) The signatory of this report declare that the recommendations reflect solely and exclusively their personal analyses and opinions, which have been produced independently, including in relation to XP Investimentos and which are subject to modifications without notice due to changes in market conditions, and that their remuneration are indirectly affected by revenue from business and financial transactions carried out by XP Investimentos.

5) The analyst responsible for the content of this report and the compliance with CVM Resolution 20/2021 is indicated above, and, in the event of an indication of another analyst in the report, the person responsible will be the first accredited analyst to be mentioned in Report.

6) XP Investimentos' analysts are obligated to comply with all the rules laid down in the APIMEC's conduct code for the securities analyst and XP Investimentos' analyst of securities conduct policy.

7) Customer service is carried out by XP Investimentos employees or by autonomous investment agents who perform their activities through XP, in accordance with CVM Resolution 16/2021, which are registered in the national association of brokers and distributors of securities ("ANCORD"). The autonomous agent of investment may not provide consulting, administration or management of customer net worth, and must act as an intermediary and request prior authorization from the client for the realization of any operation in the capital market.

8) The products presented in this report may not be suitable for all types of customer. Before making any decisions, customers should ensure that the products presented are suitable for their investor profile. This material does not suggest any change of portfolio, but only guidance on products suitable to a certain investor profile.

9) The profitability of financial products may present variations and their price or value may increase or decrease in a short period of time. Past performance is not necessarily indicative of future results. Performance disclosed is not net of any applicable taxes. The information present in this material is based on simulations and the actual results may be significantly different.

10) This report is intended exclusively for to the XP Investimentos' network, including independent XP agents and XP customers, and may also be released on XP's website. It is prohibited to reproduce or redistribute this report to any person, in whole or in part, whatever the purpose, without the prior express consent of XP Investimentos.

11) XP Investimentos' ombudsman has the mission to serve as a contact channel whenever customers who do not feel satisfied with the solutions given by the company to their problems. The contact can be made via telephone 0800 722 3710 if you are in Brazil or via ombudsman form if you are in other localities: ttps://institucional.xpi.com.br/ouvidoria.aspx/.

12) The cost of the transactions billing policies are defined in the operational cost tables which are made available on XP Investimentos website: www.xpi.com.br.

13) XP Investimentos is exempt from any liability for any damages, direct or indirect, that come from the use of this report or its contents.

14) Technical analysis and fundamental analysis follow different methodologies. Technical analysis is performed following concepts such as trends, support, resistance, candles, volume, and moving averages, amongst others. Fundamental analysis uses as information the results disseminated by the issuing companies and their projections. In this way, the opinions of fundamental analysts, who seek the best returns given the market conditions, the macroeconomic scenario and the specific events of the company and the sector, may differ from the opinions of technical analysts, which aim to Identify the most likely movements on asset prices, using "stops" limit possible losses.

15) Investment in stocks is appropriate for moderate and aggressive profile investors, according to the suitability policy practiced by XP Investimentos. Equity investments available are portion a company's capital that is traded on the market. Stock is a variable financial investment (i.e. an investment in which profitability is not pre-established and varies depending on market quotations). Investment in stock is a high-risk investment and past performance is not necessarily indicative of future results and no statement or warranty, expressed or implied, is made in this material in relation to future performance. Market conditions, macroeconomic scenario, company and sector specific events can affect investment performance and may even result in significant asset losses. The recommended duration for equity investments is medium-long term. There is no guarantee of investment return for customers' investments in stock.

16) Investment in options is preferably indicated for aggressive profile investors, according to the suitability policy practiced by XP Investimentos. In options market, the purchase or sale rights of a good shall be negotiated at a price fixed at a future date, and the purchaser of the negotiated duty should pay a premium to the seller as in a secure agreement. Operations with these derivatives are considered very high risk for presenting high risk and return relationships and some positions present the possibility of losses higher than the capital invested. The recommended duration for the investment is short-term and the customer's assets are not guaranteed in this type of product.

17) Investment in terms is indicated for aggressive profile investors, according to the suitability policy practiced by XP Investimentos. They are contracts for the purchase or sale of a certain number of shares at a fixed price for settlement within a specified period. The term of the contract is freely chosen by the investors, complying with the minimum period of 16 days and a maximum of 999 days. The price will be the value of the added share of a portion corresponding to the interest-which are set freely on the market, depending on the term of the contract. Every transaction in the term requires a guarantee deposit. These guarantees are provided in two forms: coverage or margin.

18) Investments in futures markets are subject to significant loss of principal, and are therefore appropriate for aggressive profile investors, according to the suitability policies practiced by XP Investimentos. A commodity is an object or price determinant of a future contract or other derivative instrument, which may substantiate an index, a fee, a movable value or a physical product. Commodities are considered high risk investments, which include the possibility of price fluctuation due to the use of financial leverage. The recommended duration for commodity investments is short-term and customers' assets are not guaranteed in this type of product. Market conditions and the macroeconomic scenario can affect the performance investments.

19) This institution is adhering ANBIMA Code of Regulation and best practices for the distribution activity of retail investment products.

20) XP Investments US, LLC, a broker-dealer registered with the U.S. Securities and Exchange Commission, has assumed responsibility for this research for purposes of U.S. law. All transactions arising from this research should be directed to XP Investments US, LLC, at +1 646-664-0525

xp research

banco
de atacado